

Survey Data Quality, Weighting, and Health Care Access

A NHANES 2017–2018 Analysis of Insurance, Usual Source of Care, and Self-Rated Health

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Abstract

Public-use survey files often look simple after download, but the analytic work depends on careful joining, codebook review, missing-data checks, and design-aware estimation. This study used the 2017–2018 National Health and Nutrition Examination Survey (NHANES) to examine whether health insurance coverage and having a usual source of care were associated with fair or poor self-rated health among U.S. adults. Three public SAS transport files were joined at the respondent level: demographics and survey design variables, hospital utilization and access to care, and health insurance. The analytic file contained 5,856 adults aged 18 years or older, and the survey-weighted model used 5,532 adults after excluding nonresponse on core categorical fields and imputing missing family income to poverty ratio.

In weighted descriptive estimates, 24.20% of uninsured adults reported fair or poor health compared with 17.87% of insured adults. Adults without a usual source of care had a lower weighted prevalence of fair or poor health than those with a usual source of care, 16.40% versus 19.25%, which is consistent with the idea that healthy adults may be less likely to maintain a usual site of care. In the adjusted survey-weighted model, uninsured status was not independently associated with fair or poor health, odds ratio 1.002 with 95% CI 0.774 to 1.296. Family income to poverty ratio was strongly associated with lower odds of fair or poor health, odds ratio 0.745 per unit increase with 95% CI 0.691 to 0.804. These results show why survey-weighted descriptive estimates and adjusted models should be read together rather than treated as interchangeable summaries.

1 Introduction

Health care access measures are often used as indicators of whether adults can enter and remain connected to the health system. Insurance coverage lowers financial barriers to care, while having a usual source of care can make it easier to obtain advice, preventive services, and follow-up when new problems arise. Self-rated health is a compact outcome, but it captures a respondent’s broad view of physical functioning, disease burden, and overall well-being. Because the measure is collected in many national surveys, it is useful for comparing access-related patterns across groups.

This project focuses less on causal claims and more on the analytic workflow needed to produce credible survey evidence. NHANES is a complex, multistage probability survey, and the public files are released as component-level SAS transport files that must be merged by respondent sequence number [4]. The demographics file supplies the interview weights, masked strata, and masked primary sampling units needed for variance estimation [1]. The health care access file supplies self-rated health and usual source of care fields [2]. The health insurance file supplies coverage indicators and insurance type fields [3].

The NHANES documentation emphasizes that analysts should use the appropriate sample weight and account for the complex design when estimating national quantities [5]. This matters because ignoring weights, strata, and clustering can produce misleading standard errors and, in some cases, misleading point estimates. Survey methodology texts make the same point in a broader health-survey setting, especially for public-use

health surveys where design variables are masked but still analytically required [6]. The R survey package implements these design-aware estimators and model routines for complex samples [7].

The practical question in this report is straightforward: do insurance coverage and usual source of care separate adults with fair or poor self-rated health from those reporting excellent, very good, or good health? A useful answer needs more than a two-by-two table. Coverage, usual care, age, education, race/ethnicity, and income are related to one another, so this report combines weighted descriptive estimates with an adjusted survey-weighted logistic regression model [10]. Missing family income to poverty ratio was handled with a simple predictive mean matching imputation step, with complete-case results used as a sensitivity check. The missing-data approach follows common guidance that imputation and complete-case analyses should be documented rather than hidden inside a preprocessing step [9, 8].

2 Research Questions and Hypotheses

The first research question asked whether uninsured adults had a higher weighted prevalence of fair or poor self-rated health than insured adults. The working hypothesis was that uninsured adults would report fair or poor health more often, reflecting the connection between coverage, socioeconomic risk, and health status.

The second research question asked whether adults without a usual source of care had a higher weighted prevalence of fair or poor health than adults with one or more usual places for care. The working hypothesis was that lacking a usual source of care would be associated with worse self-rated health. This hypothesis was plausible but not guaranteed, because some adults do not maintain a usual source of care precisely because they are younger or healthier.

The third research question asked whether any observed access-health differences remained after adjustment for age group, gender, race/ethnicity, education, and family income to poverty ratio. The adjusted model was intended to separate descriptive access patterns from differences that are more directly connected to demographic and socioeconomic composition.

3 Data

The analysis used NHANES 2017–2018 public-use files. NHANES releases data in multiple component files, which allows analysts to select only the variables needed for a study but also requires careful merging and codebook review. Three files were used: DEMO_J, HUQ_J, and HIQ_J. Each file contained 9,254 respondent records before the adult restriction.

Table 1: Input files and row counts

File	Content	Rows
DEMO_J	Demographics, weights, strata, PSUs	9,254
HUQ_J	Health status and access to care	9,254
HIQ_J	Health insurance	9,254

The person-level merge used SEQN, the respondent sequence number. Adults were defined as respondents aged 18 years or older at screening. The primary outcome was a binary indicator for fair or poor self-rated health based on HUQ010. Responses of excellent, very good, or good were coded as 0, while fair or poor was coded as 1. Refused and don’t know responses were set to missing.

The first exposure was current health insurance coverage from HIQ011, coded as insured versus uninsured. The second exposure was usual source of care from HUQ030, coded as having a usual place if respondents reported one place or more than one place, and no usual place if they reported that there was no place

they usually went when sick or needing health advice. Covariates included age group, gender, race/ethnicity, adult education, and family income to poverty ratio. Interview weights, masked strata, and masked PSUs came from the demographics file.

4 Methods

4.1 Data Management and Quality Checks

The R pipeline imported each SAS transport file, joined records by respondent sequence number, derived analytic variables, and exported a processed CSV file. It also wrote a variable dictionary linking analytic fields back to source files and source variables. This kind of dictionary is useful because public-use survey projects often involve many compact variable names that are easy to confuse without documentation.

Data quality checks focused on merge completeness, adult sample size, missingness in core analytic variables, valid binary outcome coding, uniqueness of respondent identifiers, and positive survey weights. The Python companion script repeated selected checks on the processed CSV file and fit a simple train/test logistic model as a model-validation side check using scikit-learn [12]. The Python model was not used for the main inference because the R analysis handled the complex survey design. It was included as a practical QA step and as an independent check that the engineered features contained signal. The data preparation and reporting workflow used the tidyverse for tabular cleaning and figure preparation [11].

Table 2: Selected data quality checks

Check	Value
Merged person-level rows	9,254
Adult rows age 18+	5,856
Adult rows with modeled outcome	5,849
Adult rows with insurance status	5,837
Adult rows with usual care status	5,854
Adult rows missing family income to poverty ratio	835

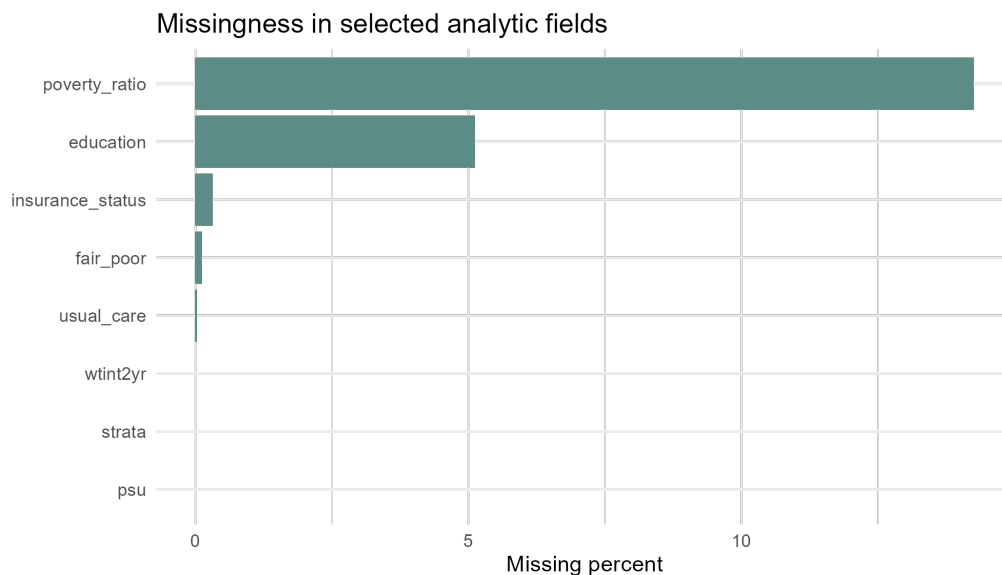


Figure 1: Missingness in selected analytic fields.

4.2 Statistical Analysis

All primary descriptive estimates used NHANES interview weights with masked strata and masked PSUs. Weighted proportions were calculated for fair or poor self-rated health overall and by insurance status, usual source of care, and age group. Confidence intervals used design-based standard errors.

The adjusted model used survey-weighted logistic regression with fair or poor self-rated health as the binary outcome. Predictors were insurance status, usual source of care, age group, gender, race/ethnicity, education, imputed family income to poverty ratio, and a missing-income indicator. Family income to poverty ratio had meaningful missingness, so predictive mean matching from the `mice` package was used to impute this continuous covariate while preserving observed distributional values [8]. A complete-case sensitivity model was run for the two primary exposure terms.

The statistical model was interpreted as associative. NHANES is cross-sectional for these fields, and the direction between health status and care-seeking cannot be established from these data alone.

5 Results

5.1 Weighted Descriptive Estimates

Overall, 18.74% of U.S. adults reported fair or poor health. The weighted prevalence was 17.87% among insured adults and 24.20% among uninsured adults. That descriptive difference supports the first hypothesis before covariate adjustment.

The usual-source-of-care comparison moved in the opposite direction from the second hypothesis. Adults who had one or more usual places for care had a weighted fair or poor health prevalence of 19.25%, compared with 16.40% among adults with no usual place for care. This pattern is plausible in a cross-sectional survey because adults in worse health may be more likely to maintain contact with the health system.

Table 3: Survey-weighted prevalence of fair or poor self-rated health

Group type	Group	Estimate	SE	95% CI low	95% CI high
Overall	Adults 18+	18.74	1.01	16.76	20.72
Insurance status	Insured	17.87	0.96	15.98	19.76
Insurance status	Uninsured	24.20	1.59	21.08	27.33
Usual source of care	Has a usual place	19.25	1.06	17.17	21.33
Usual source of care	No usual place	16.40	1.61	13.24	19.56
Age group	18–34	13.29	1.25	10.85	15.74
Age group	35–49	18.10	1.85	14.46	21.73
Age group	50–64	21.60	1.97	17.73	25.46
Age group	65+	23.01	2.26	18.59	27.43

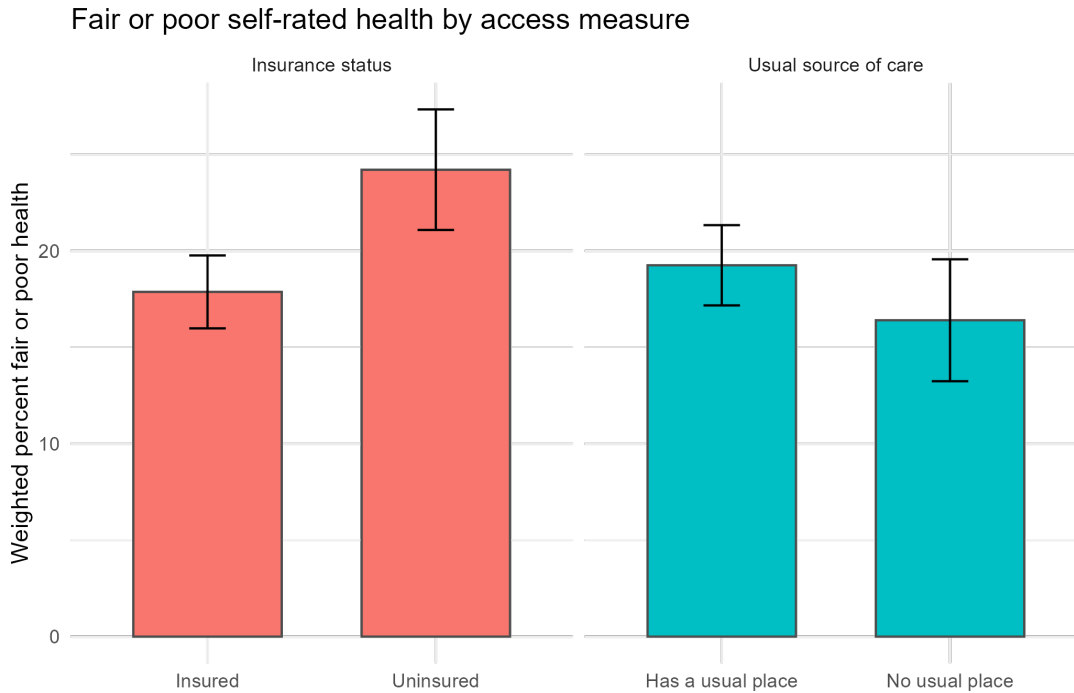


Figure 2: Survey-weighted fair or poor self-rated health by insurance status and usual source of care.

5.2 Adjusted Survey-Weighted Model

After adjustment, uninsured status was not independently associated with fair or poor health. The estimated odds ratio for uninsured adults was 1.002, with 95% CI 0.774 to 1.296. In contrast, not having a usual source of care was associated with lower adjusted odds of fair or poor health, odds ratio 0.649 with 95% CI 0.511 to 0.825. This result should not be read as evidence that lacking a usual source of care improves health. It more likely reflects selection into regular care among adults with more health needs.

Income and education showed stronger adjusted relationships with self-rated health. Each one-unit increase in family income to poverty ratio was associated with lower odds of fair or poor health, odds ratio 0.745, 95% CI 0.691 to 0.804. Adults with less than high school education had higher adjusted odds of fair or poor health than college graduates, and older age groups had higher adjusted odds than adults aged 18–34.

Table 4: Selected adjusted survey-weighted model terms

Model term	Odds ratio	95% CI low	95% CI high	p-value
Uninsured	1.002	0.774	1.296	0.989
No usual source of care	0.649	0.511	0.825	0.002
Age 35–49	1.501	1.061	2.123	0.025
Age 50–64	2.045	1.365	3.064	0.002
Age 65+	2.229	1.387	3.583	0.003
Less than high school	3.282	1.930	5.581	0.000
Family income to poverty ratio	0.745	0.691	0.804	0.000

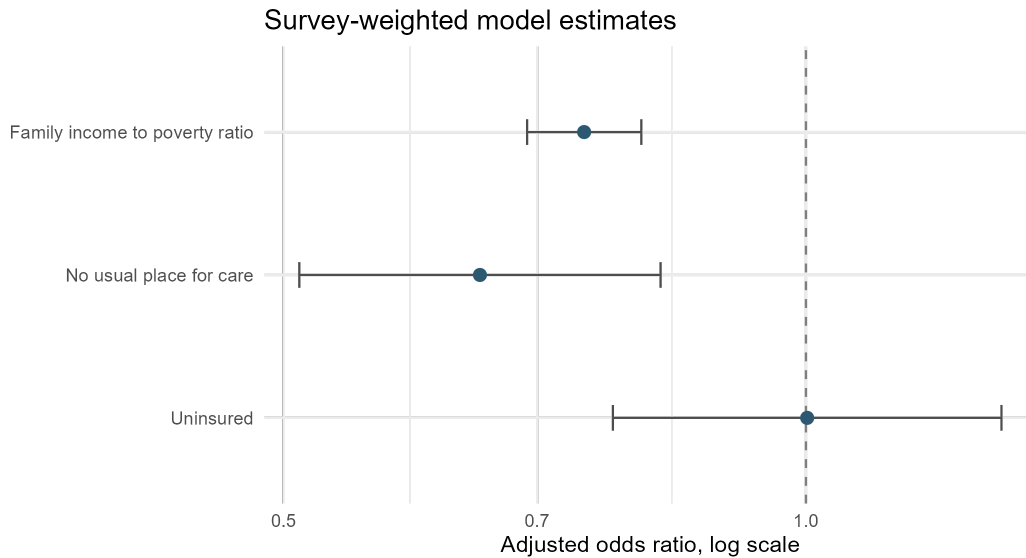


Figure 3: Adjusted odds ratios for selected survey-weighted model terms.

5.3 Missing-Data and Python Validation Checks

Family income to poverty ratio was missing for 835 adult records and for 770 records in the model-ready file. Predictive mean matching imputed this field while leaving the observed binary outcome and categorical predictors unchanged. The observed mean family income to poverty ratio was 2.559, and the mean after imputation was 2.535, suggesting that the imputation step did not materially shift the central tendency of the covariate.

The complete-case sensitivity model gave similar conclusions for the two access terms. Uninsured status remained not statistically distinguishable from insured status, odds ratio 0.891 with 95% CI 0.693 to 1.147. No usual source of care remained associated with lower adjusted odds of fair or poor health, odds ratio 0.630 with 95% CI 0.481 to 0.824. The Python validation model produced a test-set ROC AUC of 0.7834 and average precision of 0.5418, which indicated that the engineered variables had moderate predictive signal in a conventional machine-learning split. Because that Python model did not use survey design variables for national inference, it was treated as a validation side check rather than the primary statistical analysis.

6 Discussion

The descriptive results support a familiar access pattern: uninsured adults had a higher weighted prevalence of fair or poor self-rated health than insured adults. That difference did not remain after adjustment for age, race/ethnicity, education, income, and usual source of care. This does not mean insurance is unimportant. Rather, it shows that in this cross-sectional file, insurance status overlaps with other social and demographic characteristics that are also associated with health status.

The usual-source-of-care finding is a useful reminder that access variables do not always behave like simple risk markers. Adults with fair or poor health may have more contact with clinicians and may be more likely to identify a place where they usually receive care. Adults without a usual source of care may include people who are younger, healthier, or less engaged with the health system. A causal interpretation would require longitudinal information, timing, and a design that could address selection into care.

From a data quality perspective, the project highlights several points that matter in public survey analysis. First, the merge key was straightforward, but the analytic meaning of each field depended on codebook review. Second, special response values such as refused and don't know needed to be handled before modeling. Third, the survey weights and masked design variables were essential for national estimates and standard errors. Fourth, missing family income to poverty ratio was large enough to document and handle explicitly rather than silently dropping all records with missing income.

The project also shows how R, Python, and SAS can occupy different places in the same survey workflow. R handled the design-aware statistical analysis. Python provided independent processed-file checks and a train/test predictive validation layer. The SAS companion script showed how the same SAS transport inputs can be imported, recoded, and analyzed with survey procedures in a SAS environment. That separation keeps the code readable while still reflecting the multi-tool workflows common in applied survey and policy research.

7 Limitations

The analysis is cross-sectional. Insurance, usual source of care, and self-rated health were measured in the same survey cycle, so associations should not be interpreted as effects. Self-rated health is also a broad outcome and does not identify whether respondents had specific clinical conditions, unmet care needs, or recent changes in insurance. The imputation step was intentionally limited to family income to poverty ratio and was used to support a transparent workflow, not to claim that all missing-data uncertainty was fully represented. Finally, the Python model was a secondary validation exercise and did not replace the survey-weighted estimates.

8 Conclusion

In NHANES 2017–2018, uninsured adults had a higher weighted prevalence of fair or poor self-rated health than insured adults, but that difference was attenuated after adjustment for demographic and socioeconomic covariates. Adults without a usual source of care had lower adjusted odds of fair or poor health, most likely because regular care is partly shaped by existing health needs. The broader lesson is methodological: survey-data conclusions require careful file harmonization, codebook review, missing-data handling, and design-aware estimation before results are summarized for a research or policy audience.

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